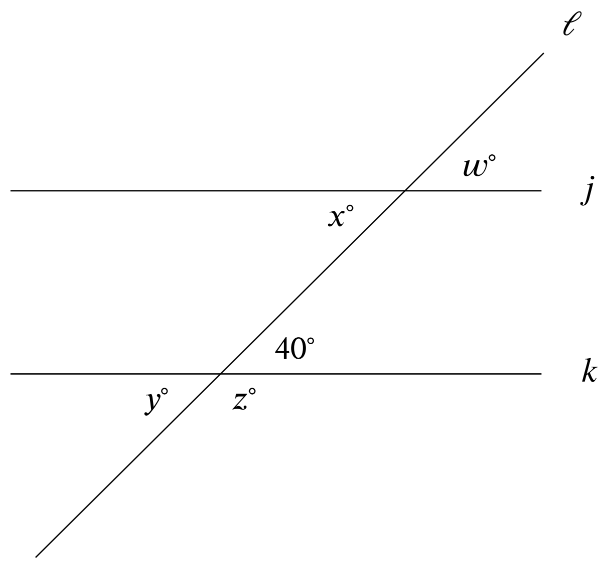
Question ID: 9d078710

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure shown, line ℓ intersects lines j and k. Which additional piece of information is sufficient to prove that lines j and k are parallel?

Answer

A. $w = 40$

B. $x = 140$

C. $y = 40$

D. $z = 140$

Correct Answer: A

Rationale

Choice A is correct. In the figure shown, lines j and k are parallel if and only if a pair of corresponding angles are congruent. It's given that one angle has a measure of $40°$ and that the corresponding angle has a measure of $w°$. Therefore, $w = 40$ is sufficient to prove that lines j and k are parallel.

Choice B is incorrect. The angle measuring $x°$ and the angle measuring $40°$ are alternate interior angles. Thus, if lines j and k are parallel, x is equal to 40, not 140.

Choice C is incorrect. The angle measuring $y°$ and the angle measuring $40°$ are vertical angles. Thus, $y = 40$, whether lines j and k are parallel or not.

Choice D is incorrect. The angle measuring $z°$ is supplementary to the angle measuring $40°$. Thus, $z = 180 - 40$, or $z = 140$, whether lines j and k are parallel or not.